

Amazon Web Services (AWS)

Cloud Computing Notes — Knowledge Base

1. Overview

AWS is Amazon's cloud computing platform, offering on-demand infrastructure and platform services (compute, storage, databases, networking, ML, and more) on a pay-as-you-go basis. It is the largest cloud provider by market share.

2. Core Services by Category

Category	Key Services
Compute	EC2, Lambda, ECS, EKS, Fargate
Storage	S3, EBS, EFS, Glacier
Database	RDS, DynamoDB, Aurora, ElastiCache
Networking	VPC, Route 53, CloudFront, API Gateway
Identity/Security	IAM, Cognito, KMS, Secrets Manager
DevOps	CloudFormation, CodePipeline, CloudWatch

3. Core Concepts

- Regions and Availability Zones (AZs) — isolated data center clusters used for redundancy and low-latency access.
- IAM (Identity and Access Management) — controls who/what can access resources, following least-privilege principles.
- VPC (Virtual Private Cloud) — an isolated virtual network for your resources.
- Elasticity — automatic scaling of compute resources (Auto Scaling Groups) to match demand.

4. Compute Options Compared

- EC2 — virtual machines; full control over the OS, most flexible, most operational overhead.
- Lambda — serverless functions; pay only per execution, no server management, ideal for event-driven workloads.
- ECS/EKS — container orchestration (EKS is managed Kubernetes) for microservices deployments.

5. Why Organizations Choose AWS

- Broadest and most mature service catalog among cloud providers.

- Global infrastructure footprint for low-latency, redundant deployments.
- Strong ecosystem of third-party tooling and enterprise support.

6. Considerations

- Pricing can be complex; requires cost monitoring (AWS Cost Explorer/Budgets).
- Steep learning curve given the breadth of services.
- Vendor lock-in risk if heavily using proprietary managed services.